Abhimanyu Prajapati

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.62
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education, Andhra Pradesh)	2020	8.63%
Kendriya Vidyalaya	Class X (Central Board of Secondary Education)	2018	71.20%

ADDITIONAL QUALIFICATION

B.S. in Data Science and Applications from Indian Institute of Technology Madras

- Advanced Certificate in Machine Learning and Data Science
- Advanced Certificate in Programming and Application Development

MASTER'S THESIS

RadXAI: Probing and Mitigating Hallucination in Radiology VLMs with Visual Retrieval-Augmented Generation and Policy Optimization | Github

- Built a V-RAG pipeline using **LLaVA-1.5-7B, Qwen2.5-14B, CheXzero, and FAISS**, with LoRA fine-tuning on **100K+ CheXpert Plus image-report pairs**.
- **Reduced hallucination rate from 45.1% to 15.6%** and improved finding-level **precision from 0.549 to 0.844**, achieving **MCC of +0.472**.
- Developed a **13K+ question hallucination benchmark** across CheXpert, MIMIC-CXR, and IU X-Ray, evaluating **Precision, Recall, F1, and MCC**.

EXPERIENCE

Research Designs & Standards Organisation (RDSO), Ministry of Railways | Software Development Intern

- Built a **Django based e-commerce platform** from scratch with **4 relational models** and **12 REST-style routes** across product, cart, checkout, and auth modules.
- Implemented a **3-stage authentication flow** signup with email activation, login/logout, and token-based password reset using Django's auth framework.
- Integrated the **PayTm payment gateway** with checksum-based signature verification to ensure secure, tamper-proof transaction confirmation.

PROJECTS

Multithread HTTP Reverse Proxy : System Programming | Github

- Built a **multithread HTTP/1.1 reverse proxy** using **sockets** and a fixed-size **thread pool** to concurrently forward **GET and POST requests**.
- Implemented a **thread-safe LRU cache** with **TTL expiration** and **HTTP cache-control policies** to reduce repeated backend requests.
- Added **HTTP parsing, partial I/O handling, HTTP error responses, synchronized logging, and atomic performance metrics**.

VacciGuard : Cloud Computing | Github

- Built an IoT streaming pipeline on **Amazon EKS** using **Kafka, Spark Structured Streaming, Redis, and S3/Parquet**.
- Reduced spike latency from **231.78s to 2.10s (110x)** by separating Redis and S3 processing into independent services.
- Cut failure recovery by **76% (64.4s to 15.67s)** using **Spark checkpointing**; implemented monitoring with **Prometheus/Grafana** and infrastructure with **Terraform**.

TransactiWar : Secure Web Application | Github

- Built a secure web app **(PHP, MySQL, Docker)** from scratch with **CSRF** protection, hardened **CSP** headers, and secure session handling no security frameworks used.
- Added **rate limiting**, parameterized queries, and **activity logging** (user, IP, timestamp) defended successfully in a live adversarial **"War Game"** against 4 other teams.
- Containerized via **Docker Compose** with network isolation and healthchecks for deployment.

FinSight-AI : RAG Evaluation Framework | Github

- Built a **RAG system for SEC 10-K filings across 5 companies** using **LangChain, FAISS, and Mistral 3 14B**.
- Improved retrieval using **hybrid BM25-FAISS search, Parent Document Retrieval, and cross-encoder reranking**, achieving **0.725 recall and 0.751 precision**.
- Achieved **0.831 faithfulness and 0.745 answer relevance** using **RAGAS.**

SKILLS

- **Programming Languages:** Python, C++, C, SQL
- **ML/DL Frameworks:** PyTorch, Transformers, Scikit-learn, OpenCV
- **Data & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **GenAI & Agentic AI:** LLM APIs, RAG, LangChain, LoRA
- **Tools & Platforms:** Git, GitHub, VS Code, Linux/Bash
- **Web & Backend:** React, Node.js, Express, PostgreSQL, Django
- **Cloud & Systems:** AWS EKS, Kafka, Spark, Terraform, Docker, POSIX Sockets

RELEVANT COURSES

- **Computer Science :** Advanced Data Structures and Algorithms, Database Management Systems, Operating Systems, Advanced Computer Networks, Object-Oriented Programming System

- **Machine Learning :** Deep Learning, Foundations of Machine Learning, Visual Computing, Representation Learning

POSITION OF RESPONSIBILITY

Techno Talks: An Interaction with a Startup Leader | Organizer | CSJMU Kanpur

- Coordinated speaker logistics and **end-to-end execution** for a **100+** student startup-industry event.

ACHIEVEMENTS

- Selected among **3,000** students from **1,34,421+** registrations (**~2.2%** selection rate) for **Amazon ML Summer School** 2026 by Amazon.
- Selected for the Internal Hackathon round of **Smart India Hackathon (SIH) 2023**.